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A Realization Method of Rotating Speed Measurement Based on Different M/T

Jingbo Xu[1,a)], Xianyu Yin[1,b)], Xiaomeng Cui[1,c)] and Zhiliang Liu[2,d)]

[1]*School of Measure-control Technology & Communication Engineering, Harbin University of Science and Technology, Harbin, Heilongjiang 150080,China;*
[2] *State Key Laboratory of Hydropower Equipment, Harbin Institute of Large Electrical Machinery, Harbin, Heilongjiang 150046, China*

[a)]Corresponding author: hitxjb@126.com
[b)]502170092@qq.com
[c)]cuixiaom@126.com
[d)]xiaozhi_231@163.com

Abstract. A realization method for rotating speed measurement is put forward in this paper. Timer 1 of STM32F103 is used to count the external pulses and its timer 2 is used to count the internal clock pulses. Two timers are set in master-slaver synchronous mode, and the rotating speed is computed based on different M/T method. This realization method can ensure the two timers strictly synchronously work, and the counting value of the master timer can also be set dynamically according to rotating speed. This method has high measurement accuracy and wide measurement range, and is easy to realize.

INTRODUCTION

Rotating speed is an important parameter of rotating machinery, and its real measurement has an important significance for monitoring the equipment and ensuring its safety operation. In general, the gear plate is mounted on the revolving axis and the speed pulses can be obtained through opto-electric sensor or magneto-electric sensor[1]. This paper adopts the different M/T method to measure rotating speed which overcomes the shortcomings of M method, T method and M/T method. A new implementation method is presented for different M/T through master-slaver synchronously timing in this paper.

PRINCIPLE OF DIFFERENT M/T METHOD

In the different M/T method, the speed pulse counter and the high frequency clock pulse counter start simultaneously when the rising edge of the speed pulse arrives. When the number of speed pulse reaches M_1, the speed pulse counter and the high frequency clock pulse counter stop at the same time[2-3]. The corresponding value of high frequency clock pulse counter is M_2. That is shown in Fig.1.Assume that the frequency of clock pulse is f_c, the corresponding counting time T_d is:

$$T_d = \frac{M_2}{f_c} \tag{1}$$

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If the tooth number of gear plate is p, the rotating speed n (r/min) can be computed as:

$$n = \frac{60M_1}{pT} = \frac{60 f_c M_1}{pM_2} \tag{2}$$

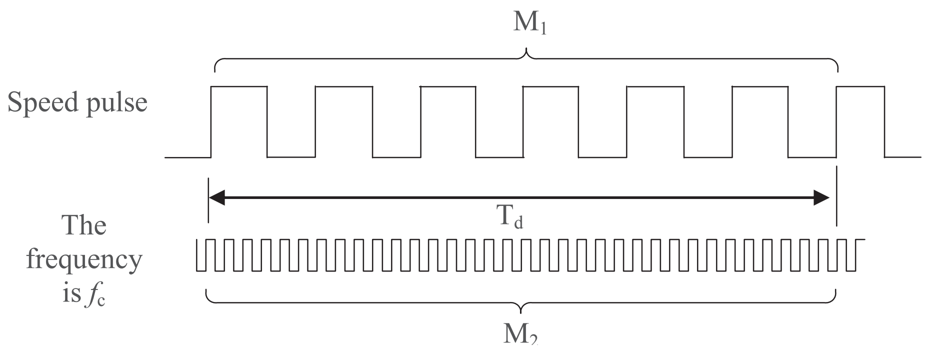

FIGURE 1. Principle of different M/T for rotating speed measurement

The value of M_1 can be adjusted dynamically according to the rotating speed. For the high rotating speed measurement, M_1 should be larger, and for low rotating speed case, M_1 should be smaller. That ensures the real-time measurement and the measurement period is also easy to set[4].

THE REALIZATION OF ROTATING SPEED MEASUREMENT

STM32F103 is taken as the processor. Its timers can be linked together internally for timer synchronization or chaining. When one timer is configured in master mode, it can start and stop the counter of another timer configured in slave mode[5-6]. As shown in Fig.2, the output TRGO1 of timer 1 links to the input trigger selection of timer 2, i.e. TRGO1 can control the operation of timer 2, which provides the conditions for simultaneous counting of M_1 and M_2.

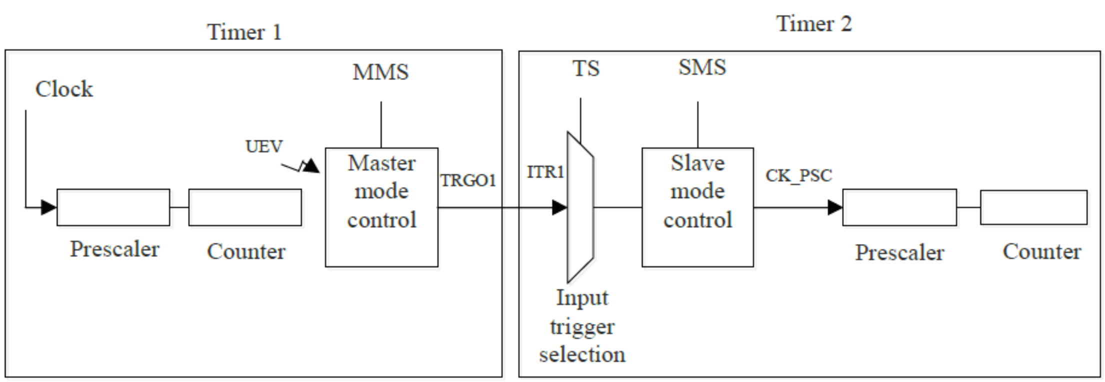

FIGURE 2. Master-Slave timing mode

Here, timer 1 is used as the master timer for counting the speed pulse and timer 2 is used as the slave timer for counting the high frequency clock pulse. Timer 1 operates in the external signal trigger mode and takes the external pulse as the clock source. Timer 1 counts up and its comparison output OC1REF, i.e. TRGO1,is linked to the input trigger selection of timer 2.Timer 2 operates in gated mode and selects ITR0 from timer 1, which is just OC1REF. The timing sequence is shown as Fig.3.

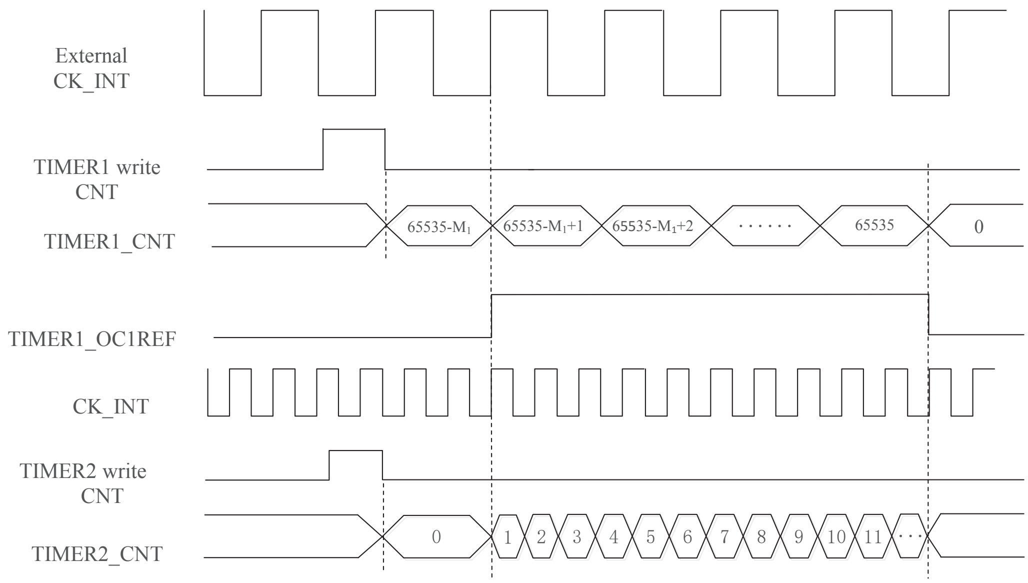

FIGURE 3. Timing sequence diagram

The initial value of the register TIM1-CNT is set to (65535- M_1) and the comparison register TIM1-CCR1 is set to (65535- M_1 +1).Timer 1 is triggered to count by the external speed pulse. When the value of TIM1-CNT is greater than or equal to the value of TIM1-CCR1, the OC1REF is high and timer 2 begins to work. That realizes the external pulse starts timer 1 and timer 2 at the same time. When the TIM1-CNT reaches 65535, it overflows and counts from zero again. In the meantime, the OC1REF is low, and timer 2 stops. Since timer 1 overflows to generate an update event, it can also be stopped in routine. Since timer 2 is stopped on the condition that timer 1 finishes M_1 speed pulses counting, the counting value M_2 of timer 2 is strictly corresponding to M_1.

The routine of rotating speed measurement is simple. Firstly, the relevant registers are initialized based on the above mentioned principle. Secondly, timer 1 and timer 2 are enabled. Finally, timer 1 will be stopped when it overflows, M_2 will be computed as follows, and the rotating speed n can be obtained according to formula 2.

$$M_2 = k \times 65536 + TIM2 \qquad (3)$$

In which, k is the number of timer 2 overflow and TIM2CNT is the value of timer 2 counting register.

M_1 should be dynamically set according to different speed. That also ensures that the high-speed and low-speed measurement cycles are basically same. Because of the rotational inertia of the rotating machine, the rotating speed can't change suddenly. M_1 can be determined according to the last measured result n' , as follows.

$$M_1 = INT(\frac{n'}{60} \times p \times T_c) \qquad (4)$$

In which, INT is the rounding function, n' is the last measured speed, and T_c is the measuring period.

EXPERIMENT

In the experiment, the number of pulses per cycle is set to 60, that is p=60. The rotating speed within the scope of 1r/min-20000r/min is tested, which covers mostly low, medium and high speed[7]. f_c is the clock pulse for timer 2 to count, which is obtained by dividing the internal clock of STM32F103. The speed measurement routine can automatically set next M_1 according to the current speed, and the testing results are shown in Table 1.

TABLE 1. Testing data

Standard Speed (r/min)	1	10	20	50	100	200
Measured Speed (r/min)	0.99996	9.99967	19.99867	49.99167	99.98000	199.96000
Relative error	-0.0400‰	-0.0330‰	-0.0665‰	-0.1666‰	-0.2000‰	-0.2000‰
Standard Speed (r/min)	500	1000	2000	5000	10000	20000
Measured Speed (r/min)	499.91668	999.93334	1999.86668	4998.33389	10000	19995.00125
Relative error	-0.1666‰	-0.0667‰	-0.0667‰	-0.3332‰	0‰	-0.2499‰

From the testing data it can be seen that this method has a wide measurement range, and high measurement precision in low and high speed cases. The maximum relative error is less than 4/10000.The dynamic setting of M_1 also ensures the real-time performance of the measurement.

CONCLUSION

In this paper, based on the principle of different M/T, a design and realization method is proposed, which makes full use of the internal resources of MCU and designs the master-slave synchronous timing mode, so that two timers can be triggered by the external pulse and cooperative work. It has high measurement precision and stability. That method relies mainly on internal timer resources, less external hardware is required, and its processing routine is also very simple, so that it has practical significance.

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REFERENCES

1. GU Yu-hai, HAN Qiu-shi, XU Xiao-li. Research on High Accuracy Rotational Speed Measurement[J]. Instrumentation Technology and Sensor,2013,10:88-91.
2. Yang Song-tao,He li-qing, An Cheng-bin. Application of DSP in high precision digital motor speed measurement[J]. Infrared and Laser Engineering,2006,10:543-548
3. Hou Tao, Fan Duo-wang. Research on Application of High Precision Digital Velocimetry Based on Incremental Photoelectric Encoder[J]. Automation & Instrumentation,2010,6: 9-11.
4. Hui Wang, Jian-tao Pu. An Improved Variable M/T Method Based on Speed Estimation for Optical Incremental Encoders[C]. Proceedings of 2012 3rd International Asia Conference on Industrial Engineering and Management Innovation (IEMI2012),2012, pp:13-20.
5. Pan Hai-hong, Chen Wei-hong, Chen Lin. Research on Synchronous Sampling Mechanism of Distributed Master-slave Multi-axis Control System[J]. Mechanical design and manufacturing,2016,8:157-160.
6. Wang Yong-hong, Xu Wei, Hao Li-ping. The Principle and Practice of STM32 Series ARM Cortex - M3 Microcontroller[M]. Beijing: Beijing University of Aeronautics and Astronautics Press, 2008.
7. Chen Jin-ying, Sun Xu, Li Can-ping. ASIC implementation of high- precision frequency measurement[J]. Journal of Chengdu University of Technology (Natural Science Edition), 2004,31 (4): 390-394.